

Independence Assessment

Is StreetLight Methodologically Independent from Iowa DOT Counts?

Overall Assessment

Bottom line:

StreetLight volumes and Iowa DOT counts should agree closely because both systems measure the same underlying traffic. High agreement alone does **not** indicate that the two are dependent.

The key issue is whether Iowa DOT counts are used to **produce** StreetLight estimates (which would create circularity), or whether they are used only for **calibration and validation** of an otherwise independent, probe-based model.

Based on available documentation and independent evaluations:

- StreetLight estimates are primarily derived from **probe/mobile/GPS data**, not directly from DOT counters
- Counts appear to be used mainly for **model calibration and accuracy checks**, not for replacing or forcing agreement at specific segments
- Reported evaluations use **holdout (out-of-sample) testing**, which reduces circularity
- External testing shows that removing training counters does **not materially change accuracy**, suggesting the model is not mechanically anchored to those sites
- It remains **unclear how much Iowa-specific count data are used during calibration**, which is the main outstanding risk to independence

Implications for Iowa DOT

- Strong agreement between DOT and StreetLight is **expected and appropriate**, not evidence of non-independence
- Correlation alone cannot diagnose dependence
- The primary risk to monitor is **local calibration or segment-level adjustment using Iowa counters**
- Full confirmation would require **vendor disclosure or Iowa-specific testing**, rather than additional statistical comparisons

Detailed Assessment

Why this question matters

When comparing StreetLight estimates to DOT/JACKALOPE observed counts, we expect **high correlation** because both systems measure true traffic volume.

High correlation is therefore **necessary but not diagnostic**.

The relevant concern is **methodological independence**:

- Independent → estimates driven primarily by probe data, with counts used only for calibration or validation
- Potentially dependent → counts directly constrain, replace, or blend into estimates at specific locations

The practical question is:

To what extent are Iowa DOT counts used to *produce* StreetLight Iowa estimates, rather than simply evaluate them?

What assessing independence entails

We focused on where and how DOT counts might enter the StreetLight estimation process.

Calibration vs. validation - Are counts used to train the model? - Are some locations held out strictly for testing?

Best practice recommends holding out sites so performance is evaluated on unseen data.

Geographic scope - National or regional calibration → lower risk of Iowa-specific circularity
- State-specific calibration → higher risk

Segment-level correction or blending - Are counted links (or nearby segments) directly adjusted to match DOT values?

This presents the greatest risk because agreement could be created “by construction.”

Leakage controls - Are nearby counters grouped during validation? - Are folds designed to avoid near-duplicates?

These steps help maintain independence between training and testing.

Evidence currently available

StreetLight documentation - Reports stratified k-fold cross-validation with held-out counters
- Indicates counts contribute to calibration
- Accuracy is evaluated out-of-sample

This is consistent with standard modeling practice rather than direct substitution of DOT data.

FHWA guidance - Complete separation between calibration and evaluation is rarely feasible
- Independent holdouts are recommended
- Some calibration use of counts is acceptable if testing remains independent

Independence is therefore treated primarily as a **validation design issue**, not absolute exclusion of counts.

Minnesota evaluation - Approximately 65 permanent counters used for training were removed from evaluation

- Accuracy changed little except at very low volumes

Interpretation: - If StreetLight were directly copying counts, accuracy would drop noticeably

- Minimal change suggests the model generalizes from probe data rather than anchoring to specific counters

What we can say with confidence

- StreetLight uses probe-derived modeled volumes
- Counts are used for calibration and validation
- Holdout testing is employed
- Independent testing suggests no mechanical anchoring to specific counters
- Strong agreement with DOT counts is therefore expected

What remains uncertain

Documentation alone cannot determine:

- how much Iowa-specific data enter calibration
- which Iowa sites or years are used
- whether Iowa evaluation locations are fully held out
- whether any segment-level blending or corrections occur

Answering these requires vendor clarification or Iowa-specific testing.

Practical takeaway

Methodological independence depends on **how the model is built and validated**, not how correlated the results appear.

Current evidence supports: - probe-driven modeling

- holdout-based evaluation

- limited risk of circularity

Further confirmation would come from: - explicit vendor documentation, or

- targeted Iowa holdout analyses